Colon & Colorectal Cancer — Biomarker Testing Intelligence

Alert Window: Last 30 Days · Generated: 2026-06-13 07:30:01

Situation Summary

Stage IV: 11 of 46
Biomarker Count: 27 entries
Top Targets: ctDNA (13) · BRAF (6) · MSI/MMR (5)
Breakthrough: 13 entries scored ≥ 4

Research & Guidelines Timeline

Biomarker testing entries ranked by significance score. Higher scores indicate greater clinical relevance. Always discuss with your oncologist.

HIGH Stage IV ctDNA VERIFIED LIQUID BIOPSY PUBLISHED MAY 26, 2026 9

Circulating tumor DNA-based assessment of disease progression after liver resection or transplantation for metastatic colorectal cancer or hepatocellular carcinoma-study protocol for a randomized, controlled trial

Circulating tumor DNA (ctDNA) monitoring is being evaluated for its clinical utility in detecting minimal residual disease (MRD) after liver resection or transplantation for metastatic colorectal cancer (CRLM) and hepatocellular carcinoma (HCC). This randomized controlled trial aims to determine if ctDNA can provide earlier detection of disease progression compared to conventional imaging and biomarkers, which have limited sensitivity.

pubmed.ncbi.nlm.nih.gov

HIGH All Stages ctDNA VERIFIED LIQUID BIOPSY PUBLISHED MAY 18, 2026 9

Circulating Tumor DNA and Precision Biomarkers in Colorectal Cancer: Implications for Diagnosis, Monitoring, and Management of Advanced Disease

Circulating tumor DNA (ctDNA) and novel precision biomarkers enhance the diagnosis, monitoring, and management of advanced colorectal cancer (CRC). Traditional markers like CEA, KRAS/NRAS, BRAF, and MSI status are inadequate for comprehensive risk stratification. The study highlights the potential of emerging biomarkers from liquid biopsies and stool-based methylation to improve individualized treatment strategies in CRC.

pubmed.ncbi.nlm.nih.gov

HIGH All Stages MSI/MMR VERIFIED LIQUID BIOPSY PUBLISHED MAY 27, 2026 7

Evaluation of Droplet Digital PCR Assay for the Detection of Microsatellite Instability in Colorectal, Gastric, and Endometrial Cancers

Droplet digital PCR (ddPCR) was evaluated for detecting microsatellite instability (MSI) using BAT-26, ACVR2A, and DEFB105A/B markers in colorectal, gastric, and endometrial cancers. The study demonstrated that ddPCR is a highly sensitive method for MSI detection, which is crucial for diagnosing Lynch syndrome and informing immunotherapy decisions.

pubmed.ncbi.nlm.nih.gov

HIGH All Stages ctDNA VERIFIED LIQUID BIOPSY PUBLISHED MAY 24, 2026 7

Practical integration of ctDNA-defined minimal residual disease in gastrointestinal cancers: testing windows and evidence-aligned management frameworks

Circulating tumor DNA (ctDNA) serves as a biomarker for minimal residual disease and tumor burden in gastrointestinal cancers, including colorectal cancer. This narrative review discusses practical testing windows, assay selection, and evidence-based management frameworks for integrating ctDNA into clinical practice. The review highlights that the strongest evidence supports ctDNA utilization in resected gastrointestinal cancers, indicating its potential role in monitoring and guiding treatment decisions.

pubmed.ncbi.nlm.nih.gov

HIGH Early Detection ctDNA VERIFIED LIQUID BIOPSY PUBLISHED JUN 12, 2026 6

Diagnostic Accuracy of Blood-Based cfDNA/ctDNA Tests for Colorectal Cancer-A Systematic Review and Meta-Analysis

Blood-based assays for colorectal cancer screening, specifically circulating tumor DNA (ctDNA) and cell-free DNA (cfDNA) tests, show variable diagnostic performance across different platforms. A systematic review and meta-analysis were conducted to assess their accuracy compared to stool-based tests. The findings highlight the need for standardized methodologies to improve the reliability of these non-invasive screening modalities in clinical practice.

pubmed.ncbi.nlm.nih.gov

HIGH Stage IV ctDNA VERIFIED LIQUID BIOPSY PUBLISHED MAY 18, 2026 6

Evaluation of ct-DNA and mutational status in liquid biopsy of patients with metastatic colorectal cancer

Plasma circulating tumor DNA (ct-DNA) levels correlate with tumor burden in metastatic colorectal cancer (CRC) patients, offering insights for personalized treatment strategies. The study underscores the utility of liquid biopsy in assessing mutational status, which may enhance clinical outcomes in CRC management.

pubmed.ncbi.nlm.nih.gov

HIGH All Stages ctDNA VERIFIED LIQUID BIOPSY PUBLISHED MAY 15, 2026 6

A novel liquid-liquid cfDNA extraction method for targeted sequencing with colorectal cancer patient samples: a pilot study

A novel liquid-liquid extraction method for circulating tumor DNA (ctDNA) from plasma was developed to enhance the quality and quantity of cell-free DNA for targeted sequencing in colorectal cancer. This pilot study addresses challenges in ctDNA mutational analysis, potentially improving non-invasive detection and monitoring of gene mutations associated with colorectal cancer.

pubmed.ncbi.nlm.nih.gov

MEDIUM Stage IV BRAF VERIFIED RESEARCH PUBLISHED JUN 04, 2026 5

Successful Management of BRAF V600E-Mutated Metastatic Colorectal Cancer With Liver Dysfunction Using Encorafenib-Based Therapy

Encorafenib-based therapy, including cetuximab and FOLFOX, is established as a first-line treatment for BRAF V600E-mutated metastatic colorectal cancer (mCRC). This case report highlights its successful application in a patient with significant liver dysfunction, addressing a gap in safety and efficacy data for this population often excluded from clinical trials.

pubmed.ncbi.nlm.nih.gov

HIGH Stage IV BRAF VERIFIED LIQUID BIOPSY PUBLISHED JUN 12, 2026 4

Clinical Impact of Baseline ctDNA RAS/BRAF Mutations on Conversion Surgery and Outcomes in First-Line Anti-EGFR Therapy for Advanced Colorectal Cancer

Baseline ctDNA RAS/BRAF mutations were assessed in patients with unresectable metastatic colorectal cancer undergoing first-line anti-EGFR therapy. The study indicates that the presence of these mutations may influence conversion surgery rates and overall outcomes. This highlights the importance of ctDNA testing for RAS/BRAF mutations in guiding treatment decisions and predicting surgical candidacy in advanced colorectal cancer.

pubmed.ncbi.nlm.nih.gov

HIGH Stage III BROAD SCOPE VERIFIED LIQUID BIOPSY PUBLISHED JUN 12, 2026 4

An exploratory multimodal pipeline for recurrence prediction in CRC with XELOX therapy

A multimodal machine learning pipeline was assessed for predicting recurrence in 86 colorectal cancer patients treated with XELOX. The study integrated clinicopathological and liquid biopsy data, identifying 17 recurrences. This exploratory research suggests that combining these data types may enhance risk stratification in CRC recurrence post-surgery and adjuvant therapy.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages BROAD SCOPE VERIFIED RESEARCH PUBLISHED JUN 08, 2026 3

Neural cues differentially modulate colorectal cancer cell behavior depending on patients' genomic background

Neural cues influence colorectal cancer cell behavior variably based on genomic background, including microsatellite instability (MSI) and KRAS/BRAF mutations. In patient-derived CRC cell lines, most neural signals enhanced clonogenicity, with specific adrenergic signals showing differential effects. This suggests that the interaction between neural signals and genomic status may inform therapeutic strategies and prognostic assessments in CRC management.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages MSI/MMR VERIFIED RESEARCH PUBLISHED JUN 05, 2026 3

Absence of co-occurrence between HER2 amplification and dMMR/MSI in a large multicentric colorectal cancer cohort

In a large multicentric colorectal cancer cohort, the study found no co-occurrence between HER2 amplification and mismatch repair deficiency (dMMR)/microsatellite instability (MSI). This suggests that HER2 amplification may be mutually exclusive with dMMR/MSI, which has implications for targeted therapy and the assessment of Lynch Syndrome. The findings highlight the need for careful biomarker evaluation in CRC to guide treatment decisions.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages BROAD SCOPE VERIFIED RESEARCH PUBLISHED JUN 05, 2026 3

Intermuscular adipose tissue as a prognostic biomarker for overall survival in colorectal cancer

Intermuscular adipose tissue (IMAT) serves as an independent prognostic biomarker for overall survival in colorectal cancer (CRC). The IMAT model may improve prognostic accuracy and inform personalized treatment strategies for CRC patients.

pubmed.ncbi.nlm.nih.gov

HIGH Stage III ctDNA VERIFIED LIQUID BIOPSY PUBLISHED JUN 01, 2026 3

Chemotherapy for patients with circulating tumour DNA-positive, stage II colon cancer (CIRCULATE)-an AIO/ABCSG trial

In the AIO/ABCSG trial, adjuvant chemotherapy in circulating tumor DNA (ctDNA)-positive stage II colon cancer patients did not meet the primary intention-to-treat endpoint, likely due to premature trial closure. However, per-protocol analysis indicates a potential benefit from adjuvant therapy in this subgroup, supporting the use of ctDNA testing for guiding adjuvant treatment decisions.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages BROAD SCOPE VERIFIED RESEARCH PUBLISHED MAY 27, 2026 3

Immune microenvironment evolution across the serrated neoplasia pathway and its relevance to immunotherapy

Serrated neoplasia pathway tumors, comprising 15%-30% of sporadic colorectal cancers, exhibit early BRAF V600E mutations and distinct immune microenvironments. Despite the efficacy of immune checkpoint inhibitors in MSI-H/dMMR colorectal cancer, these tumors demonstrate significant molecular heterogeneity, impacting their response to immunotherapy. Understanding the immune microenvironment evolution in this context is crucial for optimizing treatment strategies.

pubmed.ncbi.nlm.nih.gov

MEDIUM | Stage IV | BRAF | VERIFIED | RESEARCH | PUBLISHED MAY 20, 2026 | 3

KRAS and BRAF mutations modify adjuvant chemotherapy outcomes in early stage colorectal cancer

KRAS and BRAF^(V600E) mutations were assessed for their impact on adjuvant chemotherapy outcomes in stage III and high-risk stage II colorectal cancer. The study analyzed patients post-curative resection with molecular profiling. Results indicate that these mutations may modify the survival benefit associated with specific adjuvant chemotherapy regimens, suggesting a need for tailored treatment approaches based on mutation status in early-stage CRC.

pubmed.ncbi.nlm.nih.gov

MEDIUM | All Stages | BRAF | VERIFIED | RESEARCH | PUBLISHED JUN 12, 2026 | 2

KRAS/NRAS/BRAF Mutations and Mismatch Repair Deficiency in Patients with Early-Onset Colorectal Cancer: A Clinicopathological Analysis

Early-onset colorectal cancer (CRC) in young Chinese patients (ages 22-45) exhibits distinct clinicopathological features. The study analyzed 239 patients to investigate associations between these features and genetic mutations, specifically KRAS, NRAS, BRAF mutations, and mismatch repair deficiency. Key findings highlight the advanced stage at diagnosis and unique challenges in managing early-onset CRC, emphasizing the need for tailored clinical approaches based on genetic profiling.

pubmed.ncbi.nlm.nih.gov

MEDIUM | Early Detection | BROAD SCOPE | VERIFIED | EARLY DETECTION | PUBLISHED JUN 12, 2026 | 2

Phenotypic heterogeneity in carriers of a pathogenic MSH2 variant: implications for the diagnosis of Lynch syndrome

Carriers of a pathogenic MSH2 variant exhibit phenotypic heterogeneity, impacting Lynch syndrome diagnosis. The study highlights the importance of microsatellite instability (MSI) and immunohistochemistry (IHC) in screening algorithms for Lynch syndrome-associated tumors, which typically show MSI and loss of MMR protein expression. Understanding this heterogeneity may refine diagnostic approaches for patients with suspected Lynch syndrome.

pubmed.ncbi.nlm.nih.gov

MEDIUM | All Stages | BROAD SCOPE | VERIFIED | RESEARCH | PUBLISHED JUN 02, 2026 | 2

Traditional serrated adenoma of the colorectum: Evolving concepts in histological classification and terminology

Traditional serrated adenomas (TSAs) are rare colorectal polyps that often harbor KRAS or BRAF mutations, indicating distinct carcinogenic pathways. KRAS-mutant TSAs are predominantly found in the distal colorectum, while BRAF-mutant TSAs are more common proximal to the transverse colon. Understanding these mutations may aid in risk stratification and management of colorectal cancer associated with TSAs.

pubmed.ncbi.nlm.nih.gov

HIGH | All Stages | BROAD SCOPE | VERIFIED | GUIDELINE | PUBLISHED MAY 18, 2026 | 2

Rectifying referrals: genetics testing is underutilized in rectal cancer patient care

MSI/MMR testing recommendations are not consistently followed at the facility level, and patient compliance with genetic referrals in rectal cancer is low. When recommendations are adhered to, patients with germline mutations experience beneficial adjustments in disease management.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages BRAF VERIFIED RESEARCH PUBLISHED JUN 12, 2026 1

Molecular Landscape and Advanced Diagnostic Technologies for BRAF Mutations in Cancer: From Quantitative PCR and ddPCR to CRISPR-Based Platforms

BRAF mutations, particularly BRAF-V600E, are significant in colorectal cancer and other malignancies, driving MAPK pathway activation and impacting treatment response. The study discusses advanced diagnostic technologies, including quantitative PCR, ddPCR, and CRISPR-based platforms, for accurate detection of BRAF alterations, underscoring their importance in molecular classification and prognostic evaluation.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages BROAD SCOPE VERIFIED GENOMIC PROFILING PUBLISHED MAY 26, 2026 1

Machine learning-based preoperative classification of colorectal cancer stage using systemic inflammatory and nutritional biomarkers

Machine learning models utilizing systemic inflammatory and nutritional biomarkers can differentiate early-stage from advanced-stage colorectal cancer preoperatively. The study suggests potential clinical utility, but emphasizes the need for larger multicenter studies with external validation to confirm these findings.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages RAS VERIFIED RESEARCH PUBLISHED MAY 25, 2026 1

RAS mutation status and immune microenvironment define distinct prognostic landscapes and predict chemotherapy benefit in pMMR colorectal cancer

RAS mutational status influences the immune microenvironment in pMMR colorectal cancer, affecting prognosis and chemotherapy response. The study suggests that combining RAS status with tumor immune microenvironment features, especially CD163, may enhance patient selection for chemotherapy benefit.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages BROAD SCOPE VERIFIED GENOMIC PROFILING PUBLISHED MAY 22, 2026 1

Towards quality control and harmonization of deep learning CT radiomics: An in-silico feasibility study with virtual colorectal liver metastases

Deep learning (DL)-based radiomic biomarker prediction for colorectal liver metastases showed improved reproducibility with image harmonization and provided a reliability metric through uncertainty estimation in a controlled in-silico study. While these findings indicate methodological feasibility, clinical validation on actual CT data is necessary for translation into practice.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages KRAS VERIFIED RESEARCH PUBLISHED MAY 21, 2026 1

Frequent Detection of *KRAS*-G12C and *PIK3CA*-Q546K Mutations in MAP Tumors Highlights their Role in *MUTYH* Variants of Uncertain Significance Reclassification

KRAS-G12C and PIK3CA-Q546K mutations are frequently detected in colorectal cancers associated with biallelic pathogenic variants in MUTYH, contrasting with their rarity in sporadic cases. This enrichment suggests that these mutations may aid in the reclassification of MUTYH variants of uncertain significance. The study highlights the potential role of these specific biomarkers in the context of MUTYH-associated polyposis.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages BROAD SCOPE VERIFIED GENOMIC PROFILING PUBLISHED MAY 20, 2026 1

NOP56 Drives Colorectal Cancer Progression by Modulating p53 Acetylation through SIRT1/p300

Elevated NOP56 expression in colorectal cancer (CRC) correlates with disease progression. The study utilized TCGA datasets, tissue microarrays, and next-generation sequencing to assess NOP56's role in CRC, revealing its involvement in modulating p53 acetylation via the SIRT1/p300 pathway. These findings suggest that NOP56 may serve as a potential biomarker for CRC progression and a target for therapeutic intervention.

pubmed.ncbi.nlm.nih.gov

MEDIUM Stage III MSI/MMR VERIFIED RESEARCH PUBLISHED MAY 15, 2026 1

Preoperative MMR testing in colorectal cancer: Large-scale cohort confirms reliability but identifies tumor heterogeneity as a challenge

Preoperative biopsy for MMR assessment in colorectal cancer demonstrates high reliability in a large cohort. However, tumor heterogeneity, including weak staining and intraglandular variability, poses challenges for accurate interpretation. Implementing refined criteria can enhance the evaluation of atypical MMR patterns. Neoadjuvant therapy does not significantly affect the reliability of MMR interpretations, though clinicians should be aware of rare therapy-associated alterations.

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Appendix A — Patient-Friendly Summary

Plain-language overview for patients, caregivers, and family members

What's happening

Biomarker testing — also called molecular profiling or genomic testing — analyzes your tumor's DNA to identify specific mutations. The results determine which FDA-approved treatments may be available to you. The entries below represent the highest-scoring study per biomarker target drawn from the full research archive — not limited to the current report window. Some entries may predate the timeline above; they are included because they remain the most clinically significant reference available for that biomarker.

KRAS

Circulating tumor DNA (ctDNA) clearance as an early indicator of sotorasib + panitumumab efficacy and prognosis in KRAS G12C–mutated colorectal cancer (mCRC): Results from phase 3 CodeBreaK 300.

In the phase 3 CodeBreaK 300 trial, ctDNA clearance was assessed as an early indicator of treatment efficacy and prognosis in KRAS G12C–mutated mCRC patients receiving sotorasib and panitumumab. The findings suggest that ctDNA clearance may serve as a valuable biomarker for monitoring therapeutic response and predicting outcomes in this patient population.

ctDNA

Circulating tumor DNA-based assessment of disease progression after liver resection or transplantation for metastatic colorectal cancer or hepatocellular carcinoma-study protocol for a randomized, controlled trial

Circulating tumor DNA (ctDNA) monitoring is being evaluated for its clinical utility in detecting minimal residual disease (MRD) after liver resection or transplantation for metastatic colorectal cancer (CRLM) and hepatocellular carcinoma (HCC). This randomized controlled trial aims to determine if ctDNA can provide earlier detection of disease progression compared to conventional imaging and biomarkers, which have limited sensitivity.

TP53

Clinical impact of TP53 mutation status on survival outcomes in metastatic colorectal cancer

TP53 mutations serve as an independent prognostic marker for prolonged progression-free survival in metastatic colorectal cancer. While overall survival differences were not statistically significant, the findings suggest the need for further validation in prospective studies to assess TP53's role in therapeutic stratification.

BRAF

Detection of KRAS, NRAS and BRAF Mutations in Liquid Biopsy from Patients with Colorectal Cancer

ddPCR analysis of circulating tumor DNA (ctDNA) in colorectal cancer patients reveals that it can provide complementary information for molecular diagnosis. The study highlights the utility of liquid biopsy in detecting KRAS, NRAS, and BRAF mutations, enhancing diagnostic accuracy in this patient population.

MSI/MMR

Association of progression-free survival (PFS) with microsatellite instability–high (MSI-H) or mismatch repair–deficient (dMMR) status determined by single vs dual validated companion diagnostic (CDx) assays: CheckMate-8HW post hoc analyses.

In post hoc analyses of CheckMate-8HW, progression-free survival (PFS) was assessed in relation to microsatellite instability–high (MSI-H) or mismatch repair–deficient (dMMR) status using single versus dual validated companion diagnostic (CDx) assays. The findings suggest that the method of biomarker determination may influence PFS outcomes in colorectal cancer patients.

NGS

Rapid On-Site Next-Generation Sequencing: An Alternative to Single-Gene and Send-Out Testing in Non-Small Cell Lung Cancer and Colorectal Cancer in a Community Pathology Laboratory Setting

Next-generation sequencing (NGS) using the ThermoFisher Oncomine Precision Assay (OPA) was assessed as an alternative to single-gene panels and send-out NGS for identifying actionable mutations in colorectal cancer. The study focused on turnaround time, quantity not sufficient rates, and detection of NCCN-recommended alterations, indicating that OPA may enhance efficiency in mutation identification for targeted therapies in a community pathology laboratory setting.

HER2

Comprehensive genomic landscape of ERBB2 in Chinese GI tumors: mutation-centered landscapes and precision treatment opportunities

ERBB2 alterations in colorectal cancer (CRC) and gastric cancer (GC) predominantly occur in kinase domain hotspots, influenced by tumor-specific genomic contexts. Oncogenic ERBB2 mutations correlate with higher tumor mutational burden

(TMB) and microsatellite instability-high (MSI-H) status. These findings support the clinical evaluation of mutation-selective HER2 inhibitors and their potential combination with immune checkpoint blockade for targeted therapy in CRC and GC.

RAS

Effect of sex differences on the emergence of ctDNA RAS mutations in RAS wild-type colorectal cancer

In a single-center retrospective study of 43 patients with microsatellite stable RAS and BRAF wild-type metastatic colorectal cancer, the emergence of RAS mutations during chemotherapy was analyzed using a ctDNA assay. The study aimed to identify clinicopathological factors associated with this emergence, highlighting the potential impact of sex differences on RAS mutation development in RAS wild-type mCRC.

Genomic Profiling

Integrated Molecular Profiling of Colorectal Cancer by Tumor Location: Evidence from a Real-World Cohort with Primary and Metastatic Samples

Colorectal cancer (CRC) tumor location correlates with distinct molecular profiles. Right-sided tumors exhibit dMMR, MSI-H, and elevated TMB, while left-sided tumors show increased ERBB2 alterations and class 5 APC mutations. These findings underscore the necessity of incorporating tumor location into personalized molecular diagnostics and treatment strategies for CRC patients.

EGFR

Circulating Butyrate Attenuates Cetuximab Efficacy in Colorectal Cancer Through EGFR and AMPK-Wip1 Signaling

Butyrate may serve as a predictive biomarker for treatment response in KRAS wild-type colorectal cancer, as it reduces cetuximab efficacy via EGFR and AMPK-Wip1 signaling pathways.

miRNA

Diagnostic Performance of Circulating miRNA-92a and Related Circulating miRNA Panels for Colorectal Cancer: An Updated Systematic Review and Meta-Analysis

Circulating miRNA-92a shows superior diagnostic performance as a non-invasive biomarker for early colorectal cancer detection. The systematic review and meta-analysis suggest its potential utility, though further large-scale prospective studies are needed to standardize testing protocols and confirm clinical applicability.

TMB

Plasma-TMB (pTMB) and circulating tumor fraction (ctF) dynamics as biomarkers of benefit from the addition of atezolizumab to first-line FOLFOXIRI/bevacizumab in metastatic colorectal cancer (mCRC): A translational analysis of the AtezoTRIBE study.

In the AtezoTRIBE study, plasma tumor mutational burden (pTMB) and circulating tumor fraction (ctF) were analyzed as biomarkers for the efficacy of adding atezolizumab to first-line FOLFOXIRI/bevacizumab in metastatic colorectal cancer. The study suggests that pTMB and ctF dynamics may predict treatment benefit, providing insights for personalized therapy strategies in mCRC.

Stool DNA

Enhancing non-invasive colorectal cancer screening with stool DNA methylation markers and light gradient-boosting machine learning

Stool DNA methylation markers CNRIP1, SFRP2, and VIM demonstrated high sensitivity for non-invasive colorectal cancer screening. The application of the LightGBM machine learning algorithm improved diagnostic accuracy, suggesting a viable method for early CRC detection.

Methylation

Methylation biomarkers for early detection of colorectal cancer: From molecular discovery to clinical translation and application

Key methylation biomarkers (SEPT9, BMP3, NDRG4) have been identified for early detection of colorectal cancer (CRC) through genome-wide association studies and candidate gene approaches. These biomarkers have been validated using high-throughput technologies, facilitating their transition from research to clinical application. Early screening utilizing these methylation markers may improve patient outcomes in CRC management.

IHC

Integration of deep learning and radiomic features from multiplex immunohistochemistry images for reproducible Multi-Outcome prediction in a Multi-Center study of colorectal cancer

Fused multiplex immunohistochemistry (mIHC)-derived radiomic and deep learning features provide accurate and interpretable predictions for multiple colorectal cancer outcomes. The study supports the integration of these features into precision oncology workflows, enhancing the potential for reproducible multi-outcome predictions across different centers.

What this means for you

If you have been diagnosed with colorectal cancer, ask your oncologist: *"Has my tumor been tested for all the biomarkers that have an FDA-approved treatment?"* This is especially important if you have been diagnosed with metastatic (Stage IV) colorectal cancer, where biomarker results directly determine which targeted therapies and immunotherapies are available to you. Based on the research in this report, the most actively studied biomarkers right now are **KRAS, ctDNA, TP53, BRAF, and MSI/MMR**. Testing should happen at diagnosis — before treatment begins — because results take 1–2 weeks and can shape your first treatment decision.

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This is not medical advice. Always speak with your oncologist before making any treatment decisions.

Appendix B — Colon & Colorectal Cancer: Biomarker Testing Reference

What Is Biomarker Testing?

Biomarker testing — also called molecular profiling or genomic testing — analyzes your tumor's DNA to identify specific mutations. A single next-generation sequencing (NGS) panel can check many gene mutations at once. Results typically take 1–2 weeks and should be obtained at diagnosis, before treatment begins.

Key Biomarkers — Test All mCRC at Diagnosis

RAS (KRAS/NRAS) · BRAF V600E · MSI/MMR status · HER2 · NTRK fusion · TMB

Testing Methods

NGS (next-generation sequencing) — checks many mutations at once from tumor tissue

Liquid biopsy / ctDNA — blood test that detects tumor DNA; useful when tissue is unavailable

IHC (immunohistochemistry) — used to test MMR status and HER2

PCR — used for specific mutation detection (e.g. MSI, KRAS)

Questions to Ask Your Oncologist

"Has my tumor been tested for all the biomarkers that have an FDA-approved treatment?"

"Was my sample taken from the primary tumor or a metastatic site — and does it need to be retested?"

"How long will results take, and will they affect my first treatment decision?"

"Is a liquid biopsy appropriate for my situation?"

Abbreviations

BRAF V600E — a specific gene mutation found in ~8–10% of colorectal cancers

ctDNA — circulating tumor DNA; tumor DNA fragments found in the bloodstream

dMMR — mismatch repair deficient; indicates the tumor may respond to immunotherapy

HER2 — a protein that drives growth in ~2–3% of colorectal cancers

IHC — immunohistochemistry; a lab staining method used to detect proteins in tissue

KRAS G12C — a specific gene mutation found in ~3–4% of colorectal cancers

mCRC — metastatic colorectal cancer (Stage IV)

MSI-H — microsatellite instability-high; indicates the tumor may respond to immunotherapy

NGS — next-generation sequencing; a method that checks many gene mutations at once

NTRK — a rare gene fusion found in <1% of colorectal cancers

RAS — a gene family (KRAS/NRAS) whose mutation status affects treatment selection

TMB — tumor mutational burden; a measure of how many mutations are in a tumor

Key Resources

ClinicalTrials.gov · NCCN Guidelines (nccn.org) · Colorectal Cancer Alliance (ccalliance.org) · Fight Colorectal Cancer (fightcrc.org) · ACS (cancer.org)

Colon & Colorectal Cancer — Biomarker Testing Intelligence

About This Report

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Limitations

This report does not represent a complete picture of all published research. Database coverage, feed availability, and relevance filtering affect what appears. Research findings may not yet reflect current clinical guidelines. Always verify information at the original source before making clinical decisions.

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